

Steps to Setup Raspberry Pi Video Looper for Flash Gordon (w/o WiFi)

YouTube video: <https://youtu.be/HL4eKUpZwSk>

Step 1: You will need:

- Raspberry Pi Zero WH (H for GPIO Header already soldered!) \$23.20
- Waveshare 2.8" DPI LCD Touch Screen (be sure to get DPI) \$57.59
- MicroSD card (Sandisk, 64 GB) \$13.49
- MicroSD card reader/writer \$9.98
- USB drive (64 GB) \$16.99
- GPIO 40-pin Male to Female connector cable \$12.25 (Pkg of 2)
- Micro USB male to USB female converter (known as OTG, or on-the-go) cable \$4.99
- Power cable with switch (microUSB to USB) \$8.29
- USB power supply (or old phone charger. Should be 2.5A. Mine is 2.4A and works fine) \$12.39
- Electrical outlet, \$5.00 <https://www.homedepot.ca/product/legrand-wiremold-duplex-receptacle-box-ivory/1000132142>
- Computer with internet connection (not priced; I assume you have one!)

Total Cost: \$164.17 (Canadian, approximately \$125 US)

Other things you may want/need:

- Raspberry Pi Starter Kit: \$15.99
 - Case (two parts, top and bottom)
 - Mini HDMI to HDMI connector cable
 - Heat sink
 - 2x20 pin header (not needed if Zero WH used above)
 - Micro USB male to USB female converter OTG (on-the-go) cable
 - Power cable with switch (microUSB to microUSB)
- Raspberry Pi 5V 3A power supply (substitute for power cable and phone charger above) \$17.49
- USB Male/Female Charging Extension Cable, 22AWG, 6 ft, \$13.99 (Pkg of 2)
- USB Female/Open End Cable, 22AWG, 30 cm, \$10.00 (Pkg of 2)
- Mini HDMI to HDMI connector cable \$7.99
- HDMI cable \$9.99
- USB Hub \$9.99
- USB Keyboard, mouse \$24.97

Step 2: Install the Operating System and Setup the Video Looper

Download operating system and looper software (based on DietPi/Buster):

- Go to <https://videolooper.de>, a German site with free video looping software
- Select latest version, click on it to download. Be patient! It will eventually download, but may take a long time to even start. Do not touch video_looper_v2.6.zip.download. It will soon be unzipped and replaced by video_looper_v2.6.img. This is a disk image of the operating system for the next step.
- Place MicroSD card in card reader, reader in USB port of your home computer
- Go to <https://www.raspberrypi.com/software/>
- Click on "Download for macOS" (or as appropriate). Wait for download.
- Click on downloaded file (Imager_1.8.4.dmg) to get Raspberry Pi Imager
- Double-Click on "Raspberry Pi Imager.app"
- Choose Raspberry Pi Zero, Use Custom, video_looper_v2.6.img, Untitled volume (MicroSD card)
- Next. NO to customization, YES to erase existing data, password for your computer, allow access to removable volume, wait...
- Done! MicroSD card is named boot

Changes required for best display of video:

- Edit the file /boot/video_looper.ini using a text editor
- Find the line (near the end of the file) that begins "extra_args"
- Add the command "--aspect-mode fill" to the end of this line:
- extra_args = --no-osd --audio_fifo 0.01 --video_fifo 0.01 --align center --font-size 55 --aspect-mode fill
 - This will select the center of your video, eliminating a slim margin on either side, in order to fill the entire Waveshare screen

Step 3: Install the Waveshare Display Software

Download and modify Waveshare software:

- Go to https://www.waveshare.com/wiki/Main_Page Waveshare display list
- Select LCD screen (2.8-inch DPI LCD), takes you to https://www.waveshare.com/wiki/2.8inch_DPI_LCD
- Follow instructions following “For Raspberry Pi OS Buster branch”:
 - Place MicroSD card in card reader, reader in USB port of your home computer
 - Use text editor to edit config.txt:
 - Add the following lines at end of file:
 - # Enable Waveshare 2.8-inch DPI LCD screen (Buster OS)
 - gpio=0-9=a2
 - gpio=12-17=a2
 - gpio=20-25=a2
 - dtoverlay=dpi24
 - enable_dpi_lcd=1
 - display_default_lcd=1
 - extra_transpose_buffer=2
 - dpi_group=2
 - dpi_mode=87
 - dpi_output_format=0x7F216
 - hdmi_timings=480 0 26 16 10 640 0 25 10 15 0 0 0 60 0 32000000 1
 - dtoverlay=waveshare-28dpi-3b-4b
 - dtoverlay=waveshare-28dpi-3b
 - dtoverlay=waveshare-28dpi-4b
 - PLUS THESE ADDITIONAL LINES TO ROTATE SCREEN 270 DEGREES:
 - #Rotate by 90 degrees times value (e.g., 90 x 3 = 270 degrees)
 - display_rotate=3
 - Save file. Mine said I didn’t have permission, but somehow saved anyway. If problems, try closing without saving, then open to see if it actually saved. If you still have problems, try right-click on file in Finder, select Get Info, look for Sharing and Permissions.
 - Download “2.8inch DPI LCD DTBO” and extract the dtbo files. Copy these files to the overlays directory (/boot/overlays/). <https://files.waveshare.com/wiki/2.8inc-DPI-LCD/28DPI-DTBO.zip>
 - Save and quit the MicroSD card safely, remove from home computer

Step 4: Setup and Run the Raspberry Pi with the Waveshare Screen

- Insert the MicroSD card into the Raspberry Pi
- connect the Waveshare screen to the 40-pin GPIO interface of the Raspberry Pi. Do not insert upside-down! The combined Pi plus display should form a neat little box.
- Plug USB adapter cable (micro-USB to USB) into **inner** micro-USB connector.
- Plug USB drive with at least one video file in the root directory into the USB adapter
- Plug power cable (micro-USB to USB) into **outer** micro-USB connector. Plug into power converter / phone charger. Should be 5 Volt and minimum 2.4 Amp converter. Plug this into wall outlet.
- Click power button on your power cable, if there is one, to turn on Raspberry Pi. It will take a few seconds to start.
- See that blinking cursor in the top left of the screen? That means it's working.
- Messages should begin scrolling up screen. Just ignore.
- Eventually you should see a countdown starting at 5, followed by your videos. If you see a message, "Insert USB drive with compatible movies", there is a problem with your USB drive. Check that you have compatible video formats (.avi, .mov, .mkv, .mp4, or .m4v), and they are in the root directory.

Step 5: If You've had Problems

- Check the Waveshare.com and videolooper.de websites (where you got downloads above) for additional information. DietPi.com has information on the operating system:
 - <https://videolooper.de> Help with getting the looper to work
 - [https://www.waveshare.com/wiki/2.8inch DPI LCD](https://www.waveshare.com/wiki/2.8inch_DPI_LCD) Help with getting the screen to display
 - <https://dietpi.com/docs/> Help with the operating system
 - <https://github.com/popcornmix/omxplayer#synopsis> Help with the video player

Step 6: Logging onto the Raspberry Pi and Possible Changes

- Remove USB drive from Raspberry Pi, and attach keyboard
- Hit <Esc> when it displays "Insert USB drive with compatible movies"
- Login ID is pi, password is video (you may need to hit <Enter> first to get the login prompt)
- Change password for pi (good safety measure):
 - Type passwd, and follow the instructions
- Update time zone and keyboard:
 - Type sudo dietpi-config
 - Use the four arrow keys on the keyboard to move around the screen
 - Language/Regional Options:
 - Change time zone to America/Toronto (or as appropriate)
 - Change to US keyboard (or as appropriate)
- Edit basic Raspberry Pi settings:
 - Sudo nano /boot/config.txt
 - Make edits as required
 - <Ctrl-X> to exit, <Y>es, <Enter> to save over old file
- Edit video looper instructions:
 - Sudo nano /boot/video_looper.ini
 - Make edits as required
 - Extra_args is very useful to fix video issues. It sends these commands to the OMX video player. Check out <https://github.com/popcornmix/omxplayer#synopsis> for additional useful OMXPlayer instructions. I could not get --win or --crop to work, but --aspect-mode worked to fill the screen, instead of black borders top and bottom.
 - <Ctrl-X> to exit, <Y>es, <Enter> to accept save filename
- Shutdown: sudo halt
- Reboot: sudo reboot

Step 7: Edit Your Video

- Edit your clips for a 4:3 (width: height) screen. Try more close-ups as the screen is very small! For action scenes, you may have to chase the action around the screen using, for example, “Ken Burns” style cropping.
- By default, videos will run in alphabetical order. Rename files starting with a number to control the order.
- There is a black gap between clips. To avoid this, it’s best to combine all clips into a single file.
- If your pinball is movie-themed, there are likely video clips from the film available on YouTube:
 - Go to YouTube
 - Search “Flash Gordon movie clips” (or appropriate film title)
 - Select a clip
 - Download it to your computer:
 - Pause the clip, and click on the top line URL (e.g., <https://www.youtube.com/watch...>)
 - Change the word youtube to ssyoutube, and hit enter
 - This takes you to a download screen. Click on “Download MP4 720” when it is displayed.
 - NOTE: This download method no longer appears to work. Try Googling “How to download videos from YouTube” for a current method.
 - Edit clips
 - Use your favourite video editor to cut down and combine clips, keeping only the parts you want
 - Your original movie was likely widescreen, 21.6:9, quite a bit larger than the standard video editor size of 16:9. So you will lose some of the width of the film. But then when it goes onto your Waveshare screen it is reduced again to 4:3, losing more width. Edit your clips keeping this in mind. The `extra_args` command we modified above ensures the center of your video will be displayed, eliminating a strip from the left and right. Try keeping the action right in the center. You may need to go back and re-edit a couple of times to get everything you want on the screen.
 - Just for Flash Gordon: The movie starts out with a series of panels from the comic books. But in order to fill the screen, they stretched them sideways. To cover this up they have the clips go by in rapid succession. If you slow it down, everyone looks quite fat. So to use these clips, the first thing you want to do is compress them back down. I used a free online service called <https://www.onlineconverter.com/resize-video> . Select width and Height (fixed), set to 1280 x 960. This produces a 4:3 clip that is perfect for the Waveshare. Then use your video editor to slow down the faster clips to 25% or less.
- Copy edited clip files to your USB drive, and attach it to the Raspberry Pi with the connector cable

Step 8: Install the Raspberry Pi in Your Pinball Machine

- Remove pinball plastics
- Then work with wood, Styrofoam, other materials to hold it at the perfect height within machine.
- Use soft, compressible anti-static foam material, the kind that is used to ship electronics, as the topmost layer to press and hold the display between the plastic and the support.
- My construction looked like this:
 - Base: Wood, 3 5/8" x 2 3/8" x 1/4"
 - Support block: Styrofoam (builder's choice!), 2 1/2" x 2 3/8"
 - The block is 1/2" high in the front, and 13/16" high at the back
 - Pad on top: two layers of antistatic foam, 2 1/2" x 2 3/8" x 1/4" (total height 1/2")
 - Use double-sided duct tape to hold pieces together. Block goes at right end of base, pad on top.
- Attach the construction to the playfield with screws. Raspberry Pi and display sit on top, with cables coming out upper end and working their way through the available slot into the cabinet below. The 40 GPIO soldered pins hold it in place solidly against the padding.
- Find the perfect spot for the Pi behind the window, by repeatedly putting the plastic on, observing the results, taking the plastic off, and making minor adjustments. The final spot can be marked with a bit of masking tape on the pad.
- You might need a bit of electrical tape to block the light from the window just below the Pi.
- Put the plastic in place, and screw down with plastic caps. Ming plastic is not used.
- There are two ways I can think of to power your setup. One is to run a USB extension cable up into the backbox, connecting it to a USB connector that you have soldered to an available 5-volt and ground source. I have not done this and so I can't advise you much more than this. The other way to power the setup is to install a power outlet in the cabinet by tapping into the 120-volt power lines running from the power switch on your pinball to the transformer. Both of these have their challenges.
- The USB power cable needs to be long enough to reach power within the machine, and allow easy movement of the playfield. It also needs to be a charging cable, not a data cable. 24 AWG, or even 22 is available (smaller is better).
- The Flash Gordon has a secondary power setup just for the strobe light, which you probably no longer use. It's easy to find the blue and yellow wires, straight from the power switch into the small transformer, and use them to power an outlet you can plug into.
- Unplug the machine before doing any work on the power lines!
- Cut the blue and yellow, 120-volt power lines running from the power switch to the transformer. Connect the two cut ends of the blue line to the silver connectors on the outlet, and the two end of the yellow lines to the gold connectors. Attach to side of cabinet, and add cover.

Step 9: Keep copies of everything you downloaded for this project, in case it is no longer available online when you need it!

Keep copies of:

- Imager_1.8.4.dmg
- Video_looper_v2.6.img
- 28DPI_DTBO (directory)
 - vc4-kms-DPI-28inch.dtbo
 - waveshare-28dpi-3b-4b.dtbo
 - waveshare-28dpi-3b.dtbo
 - waveshare-28dpi-4b.dtbo
 - waveshare-28dpi.dtbo
 - waveshare-touch-28dpi.dtbo
- Instructions above, including lines added to config.txt and video_looper.ini
- Any additional changes you did for your particular setup
- All your video files

Keep a copy on your home computer, and include it in your regular backup procedures.

Keep a copy on the USB drive, and maybe even the MicroSD card.

Take screen shots of these instructions. Print them out and keep them with your pinball manual!

The next owner of your pinball will thank you.

Pros and Cons of Powering off the Raspberry Pi

(As opposed to using a shutdown command)

There is an issue with turning the Raspberry Pi off without shutting it down first. Just like shutting down your computer, you don't just unplug it. Theoretically you could interrupt some important file maintenance operation and corrupt your MicroSD card. But, worst case scenario, your Pi will stop running, so you pull out the MicroSD card and repeat all the steps above.

A better way to shut it down would be to connect it to your WIFI, and get an app on your phone that can shut it down remotely. Theoretically this can be done, but I didn't get it working. I was able to get WIFI working with other operating systems, but couldn't get video looping working with any of them!

Cons

- If Pi is interrupted while writing to MicroSD, you can corrupt it, and videos will no longer run
- I will be giving up on resolving this issue. And that really bugs me.

Pros

- Very simple
- No dependency on human intervention
- No one but me will ever do anything else
- Will not feel bad when I don't bother
- Other commercial options handle it the same way. I am told there have been few problems (but not zero!)
- No WIFI connection needed
 - WIFI connection is a potential security risk
 - WIFI password not used, so no one can possibly hack your Pi remotely to access your network
 - I have not been able to get WIFI to work with this OS!
- The worst that can happen is, the MicroSD card needs to be reprogrammed by repeating all the previous steps
 - Absolute worst, maybe MicroSD needs to be replaced
 - ABSOLUTE worst, you may need to watch this video again

UPDATE: I have had the Raspberry Pi turn off and on with the pinball machine main switch, as setup in the description above, for a couple of years now. I generally try to turn it off while the video is running, not when it is booting up or between videos. I feel this minimizes any chance that the Pi is writing to the data card when it is shut down. So far it has given me no problems.